Xinbao Qiao

xinbaoqiao@cuhk.edu.hk | Homepage | GitHub | Google Scholar

Education

The Chinese University of Hong Kong — Hong Kong
PhD student in Information Engineering — 2026–present

Zhejiang University — Zhejiang
M.Sc. in Artificial Intelligence — 2022–2025

- Major GPA: 90/100; rank: 3/25.
- Selected coursework: AI Algorithms and Systems (100), Secure Artificial Intelligence (97).

Shandong University — Shandong
B.Eng. in Communication Engineering — 2018–2022

- Major GPA: 82.47/100.
- Third-class Academic Award, 2018–2019.

Research Interests

- Trustworthy Machine Learning, including interpretability, privacy, security, and robustness.
- The intersection of AI with network, including AI for communication, communication for AI.

Publications

[1] When Sample Selection Bias Precipitates Model Collapse.
Xinbao Qiao, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang, Yan Pang. *ICML, 2026.*

[2] Beyond Binary Erasure: Soft-Weighted Unlearning for Fairness and Robustness.
Xinbao Qiao, Ningning Ding, Yushi Cheng, Meng Zhang. *AAAI, 2026.*

[3] Hessian-Free Online Certified Unlearning.
Xinbao Qiao, Meng Zhang, Ming Tang, Ermin Wei. *ICLR, 2025.*

[4] DynFrs: An Efficient Framework for Machine Unlearning in Random Forest.
Shurong Wang, Zhuoyang Shen, **Xinbao Qiao**, Tongning Zhang, Meng Zhang. *ICLR, 2025.*

[5] Federated Learning as Optimal Transport: Barycentric Multi-Prototype Classification for Pretrained Model.
Xinbao Qiao, Wenjing Yan, Ying-Jun Angela Zhang. **Under review**, *NeurIPS 2026.*

[6] Illusory Pattern Perception Drives Spurious Inference in Large Language Models.
Peihua Mai, Zhuoyan Shao, **Xinbao Qiao**, Meng Zhang, Xinyue Zhou, Yan Pang. **Under review**, *NeurIPS 2026.*

[7] Learn What Matters: Data Pruning for Efficient Decentralized Learning.
Xinbao Qiao, Xunhao Jiang, Zuozhu Liu, Peng Sun, Meng Zhang. **Under review**.

Research Experience

Research on Data-Centric ML Systems
Advisor: Prof. Meng Zhang, Zhejiang University — M.Eng. Student, 03/2023 - 12/2025

- **Data Influence Attribution.** Developed theoretically guaranteed approaches to quantify and analyze the influence of data on learning models, and designed unlearning techniques to remove these influences.
- **Trade-offs in Trustworthy Artificial Intelligence.** Investigated and developed data-centric methodologies to balance fairness, robustness, privacy, and utility in machine learning systems.
- **Cloud-Edge Collaborative Human-Space Healthcare.** Architected cloud-edge collaborative frameworks for real-time multimodal data monitoring of human health using video, audio, and sensor signals.

Research on Trustworthy LLM systems
Advisor: PANG Yan, James, National University of Singapore — Full-time Research Intern, 06/2025 - 12/2025

- **Illusory Pattern Perception in LLMs.** Examined whether LLMs exhibit illusory pattern perception, a known human cognitive bias, and offered an interpretable framework to account for LLM behavior.
- **Synthetic Data Evaluation.** Developed a distributed Wasserstein collaborative algorithm, providing a private Wasserstein methodology for synthetic data evaluation when real data is limited.

Skills

Programming: Python, PyTorch, TensorFlow, MATLAB, C/C++, SQL, Linux, Docker, Git, LaTeX.
Languages: Mandarin Chinese (native), English.